INTRODUCTION TO BIOINFORMATICS

Random Forest in Bioinformatics

Gene-Expression-Based Tumour Classification

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Random Forest combines many decision trees into one prediction

- **Ensemble learning:** combines predictions from multiple trees
- **Bootstrap sampling:** each tree trains on randomly sampled observations
- **Random feature selection:** subsets of features considered at each split
- **Majority voting:** decides the final class in classification
- **Versatility:** classification and regression

Why it suits gene-expression data: many features, complex patterns, robust to noise.



How Random Forest Works

Stepwise construction from bootstrap sample extraction to aggregated ensemble prediction.

01

Bootstrap Sampling

Randomly draws training instances with replacement.

02

Tree Construction

Builds an ensemble of distinct decision trees.

03

Feature Subspace

Selects random subsets of gene features per split.

04

Individual Predictions

Each tree evaluates sample expression levels.

05

Majority Voting

Aggregates individual outputs to decide the final class.

ENSEMBLE VOTING EXAMPLE

12 vs 8

Final Vote: Malignant-like

Consider an ensemble of **20 decision trees** evaluating an unknown sample: **12 trees vote Malignant-like** while **8 trees vote Benign-like**. Through majority rule, the ensemble issues a consensus prediction of **Malignant-like**, reducing variance compared to any single constituent tree.

Gene-Expression-Based Tumour Classification

Gene-expression patterns vary between biological samples, and machine learning can help analyse them.



Gene-expression features

Benign-like

Simulated samples assigned to this class.



Random Forest classifier

Malignant-like

Simulated samples assigned to this class.



Predicted class

Objective

Demonstrate how Random Forest uses numerical gene-expression features to classify samples and how predictions are evaluated.

Callout: Educational simulation, not a clinical diagnostic system.

Simulated Gene-Expression Dataset

200 synthetically generated samples · 20 numerical gene-expression features

ILLUSTRATIVE SYNTHETIC DATA (INVENTED VALUES, NOT REAL MEASUREMENTS)

Sample	Gene 1	Gene 2	Gene 3	Target
S001	2.4	7.8	3.1	Benign-like
S002	3.0	6.9	2.8	Benign-like
S003	8.1	2.6	7.7	Malignant-like
S004	7.5	3.2	8.3	Malignant-like

Rows are samples, columns are gene features; labels drive supervised classification

Dataset Matrix Setup

- **200** synthetically generated samples
- **20** numerical gene-expression features
- **2** classes: Benign-like & Malignant-like
- Labels drive supervised classification

Full dataset will be generated in a separate Google Colab practical demonstration.

Random Forest Implementation

Tools: Python, Google Colab, Pandas and NumPy, Scikit-learn, Matplotlib and Seaborn.

WORKFLOW PIPELINE

1. Generate the simulated dataset
2. Separate features and target labels
3. Stratified train/test split
4. Train a Random Forest classifier
5. Predict labels for test samples
6. Evaluate the predictions

EVALUATION METRICS

- **Accuracy:** overall proportion of correct predictions
- **Precision:** share of positive predictions that are correct
- **Recall:** share of actual positives correctly identified
- **F1-score:** harmonic mean of precision and recall
- **Confusion matrix:** actual vs predicted classes

Full implementation is demonstrated separately in Google Colab.

Understanding Classification Results

Illustrative example — not actual experimental results.

ILLUSTRATIVE CONFUSION MATRIX (HYPOTHETICAL)

Actual / Predicted	Benign-like	Malignant-like
Benign-like	8	2
Malignant-like	1	9

17 correct classifications, 3 misclassifications

DERIVED PERFORMANCE METRICS

85%	81.8%	90%
Accuracy (17/20)	Precision (9/11)	Recall (9/10)

For this hypothetical matrix only: Accuracy = 17/20 = 85%; Malignant-like precision = 9/11 ≈ 81.8%; Malignant-like recall = 9/10 = 90%.

Accuracy alone is insufficient: precision, recall, F1-score and error patterns show what kinds of mistakes occur. Actual metrics will be calculated in the separate Google Colab demonstration.

Applications and Conclusion

BIOINFORMATICS APPLICATIONS

- Gene-expression pattern analysis
- Classification of biological samples
- Biomarker prioritisation
- Genomic data analysis
- Personalized medicine research
- Target discovery workflows

SUMMARY TAKEAWAY

Random Forest combines multiple decision trees to reliably classify complex biological profiles. While synthetic data validate algorithmic mechanics, real translational breakthroughs demand validated cohorts and rigorous scientific interpretation.

KEY REFERENCES

1. Breiman, L. (2001). *Random Forests*. Machine Learning, 45, 5–32.
2. Scikit-learn documentation: RandomForestClassifier API.
3. NCBI Gene Expression Omnibus (GEO database repository).
4. Introduction to Bioinformatics course syllabus.